

Lablet 1 Solution

Tasks

Update the configuration on **Sw1** to complete the following tasks:

Task 1: IPv4 Addressing

- Configure IPv4 addresses on Sw1 interfaces using the data from table.

Task 2: Network Appliance Connectivity

- Configure the interface to WLC01 as an 802.1Q Trunk using the data from table.
- Ensure only mgmt, guest, and user VLAN traffic transmits across the trunk.
- Ensure mgmt traffic remains untagged across the trunk.

Task 3: Static Routing

- Configure a static default route through R1.
- Use a metric of 10 to add a backup static default route through R2.
- Use the next-hop IP addresses for all static routes.

Task 4: Access Control Lists

- Create and apply an ACL numbered 101 to an interface preventing telnet traffic from Server01 to reach the WLC01 mgmt address.
 - All other traffic must be allowed.
- Ensure the ACL is only applied to a single interface.

Note: Server01 is available for testing and verification. No configuration should be changed on Server01.

Task 1 - IPv4 Configuration

Sw1

```
enable
config t
!
interface e0/0
  no shutdown
  no switchport
  ip address 10.1.3.254 255.255.255.0
interface e0/1
  no shutdown
  no switchport
  ip address 10.2.3.3 255.255.255.0
interface vlan 201
  no shutdown
  ip address 10.0.201.1 255.255.255.0
```

```
exit
end
```

Verification

Sw1

```
show ip int bri
```

Interface	IP-Address	OK?	Method	Status	Protocol
Ethernet0/0	10.1.3.254	YES	manual	up	up
Ethernet0/1	10.2.3.3	YES	manual	up	up
Ethernet0/2	unassigned	YES	unset	up	up
Ethernet0/3	unassigned	YES	unset	up	up
Port-channel1	unassigned	YES	unset	up	up
Vlan201	10.0.201.1	YES	manual	up	up

```
ping 10.1.3.1 repeat 5
```

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 10.1.3.1, timeout is 2 seconds:

.!!!!

Success rate is 80 percent (4/5), round-trip min/avg/max = 1/1/3 ms

```
ping 10.2.3.2 repeat 5
```

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 10.2.3.2, timeout is 2 seconds:

.!!!!

Task 2 - VLAN Trunk

Sw1

```
enable
config t
!
interface e0/2
  switchport trunk encap dot1q
  switchport mode trunk
  switchport trunk allowed vlan 31,99,201
  switchport trunk native vlan 201
  no shut
end
```

Verification

Sw1

```
show interface trunk
```

Port	Mode	Encapsulation	Status	Native vlan
Et0/2	on	802.1q	trunking	201
Port Et0/2	Vlans allowed on trunk 31,99,201			
Port Et0/2	Vlans allowed and active in management domain 31,99,201			

```
Port          Vlans in spanning tree forwarding state and not pruned
Et0/2         31,99,201
```

Note: This will only work if Tasks 1 and 2 are both completed successfully.

```
ping 10.0.201.11 repeat 5
```

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 10.0.201.11, timeout is 2 seconds:

```
.!!!!
```

Success rate is 80 percent (4/5), round-trip min/avg/max = 1/1/1 ms

Task 3 - Static Routing

Sw1

```
enable
config t
ip route 0.0.0.0 0.0.0.0 10.1.3.1
ip route 0.0.0.0 0.0.0.0 10.2.3.2 10
end
```

Verification

Sw1

Note: This config verification does NOT require any other task to have been completed successfully.

```
show run | inc ip route
```

```
ip route 0.0.0.0 0.0.0.0 10.1.3.1
ip route 0.0.0.0 0.0.0.0 10.2.3.2 10
```

Note: This operational verification requires that Task 1 was completed successfully.

! Check "healthy" default route

```
show ip route | begin Gateway
```

```
Gateway of last resort is 10.1.3.1 to network 0.0.0.0
```

```
S* 0.0.0.0/0 [1/0] via 10.1.3.1
```

```
10.0.0.0/8 is variably subnetted, 6 subnets, 2 masks
```

```
C 10.0.201.0/24 is directly connected, Vlan201
```

```
L 10.0.201.1/32 is directly connected, Vlan201
```

```
C 10.1.3.0/24 is directly connected, Ethernet0/0
```

```
L 10.1.3.254/32 is directly connected, Ethernet0/0
```

```
C 10.2.3.0/24 is directly connected, Ethernet0/1
```

```
L 10.2.3.3/32 is directly connected, Ethernet0/1
```

! Shutdown interface e0/0 to R1

```
enable
```

```
config t
```

```
interface e0/0
```

```
shutdown
```

```
end
```

```
! Wait 5 seconds for interface to go down
```

```
show ip route | begin Gateway
```

```
Gateway of last resort is 10.2.3.2 to network 0.0.0.0
```

```
S* 0.0.0.0/0 [10/0] via 10.2.3.2
    10.0.0.0/8 is variably subnetted, 4 subnets, 2 masks
C    10.0.201.0/24 is directly connected, Vlan201
L    10.0.201.1/32 is directly connected, Vlan201
C    10.2.3.0/24 is directly connected, Ethernet0/1
L    10.2.3.3/32 is directly connected, Ethernet0/1

! Reenable interface e0/0
enable
config t
interface e0/0
no shutdown
end
```

Task 4 - ACL

Sw1

```
enable
config t
!
access-list 101 deny tcp host 10.5.6.6 host 10.0.201.11 eq 23
access-list 101 permit ip any any
!
interface vlan 201
ip access-group 101 out
!
end
```

Verification

Sw1

```
show access-lists 101
```

```
Extended IP access list 101
  10 deny tcp host 10.5.6.6 host 10.0.201.11 eq telnet (1 match)
  20 permit ip any any (5 matches)
```

```
show ip int vlan 201 | inc access list
```

```
Outgoing Common access list is not set
Outgoing access list is 101
Inbound Common access list is not set
Inbound access list is not set
```

Note: This operational verification requires that Tasks 1, 2, & 3 were all completed successfully.

Server01

```
telnet 10.0.201.11
```

```
Trying 10.0.201.11 ...
```

```
% Destination unreachable; gateway or host down
```

```
ping 10.0.201.11 repeat 5
```

```
Type escape sequence to abort.
```

```
Sending 5, 100-byte ICMP Echos to 10.0.201.11, timeout is 2 seconds:
```

```
!!!
```

```
Success rate is 100 percent (5/5), round-trip min/avg/max = 2/2/4 ms
```